



AI4ALL Ignite Rubrics

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Project Overview

Throughout the program, you will work in a group with guidance from a Mentor and Student Coordinator on an AI/ML project, including the preparation of data and implementation of an AI/ML solution while investigating the role of data throughout the AI development cycle. Your Student Coordinator will act as your group's "project manager" by checking progress during your weekly cohort meetings. They are also available to answer technical questions and help you troubleshoot your project. Your Student Coordinators are available to answer any higher-level questions you have about your project, as well as any questions you have about a career in AI. **Your Mentor is NOT your project manager or responsible for the completion of your project.** That is your group's shared responsibility.

You and your group are encouraged to pursue supervised learning projects, especially if it is the group's first attempt at a project of this nature. You may consider pursuing a different type of project (unsupervised learning, reinforcement learning, NLP, etc.) only if all your group members and Student Coordinators agree to do so. The workshops throughout the program will focus on supervised learning.

You and your group will begin by collaborating to select a topic or challenge of interest that can be investigated using supervised learning (or another type of machine learning if the group and Mentor agree). The project is intentionally designed to be open-ended and not too prescriptive to give you the freedom to investigate any topic that is of interest to your group.

Project & Group Management

At the beginning of the program, your group will fill out the [group norms document](#) and set up a time to meet outside of your class time. Be sure to determine how often you will meet as a group, how much time to allot for those meetings, and how you will communicate with each other. Your Student Coordinator, who acts as your project manager, will have access to the group norms and will ensure they are completed. During weeks 2 & 3, your Student Coordinator will help you define the scope of your project. Every week, your Student Coordinator will check in with you during your cohort meeting and ask you the following questions:

- *What is going well or on track?*
- *Are there any impediments in your way?*



- *What do you need feedback/input on?*
- *What will you do over the next week?*

Be prepared to answer those questions every week, and be honest with your Student Coordinator so they can help you work with your group to complete the program. You can always schedule a meeting 1:1 or as a group with your Student Coordinator to address any concerns that arise outside of cohort times.



Project Proposal (Doc)

The purpose of the proposal is to inform your Student Coordinators and peers about what your group plans to do for your Final Presentations. Groups will receive feedback on their Project Proposals from Student Coordinators; other groups will share the feedback based on the proposal presentation. Groups should share their Project Proposals directly with their Student Coordinators.

Proposal requirements:

- No more than two pages
- Names of group members
- The topic of interest and summary of why this topic is interesting/important, how data will be used, and how the group can investigate the impact
- Type of machine learning algorithm(s) the group intends to use
- Research question that the group hopes to answer using data
- Dataset(s) of interest that are relevant to the topic; the dataset must have enough data to test and train a model
- Identify potential sources of bias
- Citations:
 - At least 3 to 4 citations related to the proposed topic
 - Sources of data



Project Proposal Rubric

No submission, missing names, or lack of citations are graded as a zero.

RUBRIC Project Proposal					
Criteria	5 Exceeds Requirements	4 At Times Exceeds Requirements	3 Meets Requirement	2 Beginning to Meet Requirements	1 Needs Major Improvement
Research Question	The research question is proposed; no modifications are needed.	The research question is proposed; may need slightly narrowing or modifications	The research question posed; needs significant narrowing or modifications	Research question is proposed but it is unclear what is being asked.	No research question was proposed.
Topic & Summary	The topic is interesting and requires the use of data to make decisions or predictions that would otherwise be impractical to calculate or derive. The impact is being considered.	The topic is interesting and requires the use of data to make decisions or predictions that would otherwise be impractical to calculate or derive; and may still need minor refining/narrowing. The impact is being	The topic requires the use of data to make decisions or predictions that would otherwise be impractical to calculate or derive; and needs refining/narrowing. The topic is not particularly interesting. The impact is being considered.	The proposed decisions or predictions are trivial and can be calculated or derived without AI/ML. OR The impact is not being considered.	The topic does not require the use of data.



		considered.			
Sources of bias	Identifies potential sources of bias and at least two practical ways to mitigate its negative impacts.	Identifies potential sources of bias and at least one practical way to mitigate its negative impacts.	Identifies potential sources of bias.	Identifies potential sources of bias that are not relevant to this project.	Does not identify potential sources of bias.
Dataset	Datasets(s) of interest demonstrate: Substantial research efforts and independent thought. Relevance to topic. Sufficient amount of data for training and testing.	Datasets(s) of interest demonstrate: Substantial research efforts. Relevance to topic. Sufficient amount of data for training and testing.	Datasets(s) of interest demonstrate minimal: Research efforts. Relevance to topic. Sufficient amount of data for training and testing.	Datasets(s) of interest demonstrate minimal: Research efforts. Relevance to topic. Insufficient amount of data for training AND testing.	Datasets(s) of interest are inappropriate, completely irrelevant, or absent.
Citations	At least 7 citations related to the topic of interest. Source(s) of data included.	5 to 6 citations related to the topic of interest. Source(s) of data included.	3 to 4 citations related to the topic of interest. Source(s) of data included.	2 citations related to the topic of interest. Source(s) of data included.	Fewer than 2 citations related to the topic. OR Source(s) of data not included.
Overall Score: ____/25 (Calculate the total score by summing up the scores for each criterion)					



Proposal Presentation (Slides)

Groups should share their project presentation material (slides, links, docs, etc.) in Week 7's Lesson on Ed STEM within 24 hours of Project Proposal presentations.

Presentation requirements:

- 10-minute presentation
- Share lessons learned about AI impact and ethics
- Summary of Project Proposal
 - Names of group members
 - The topic of interest
 - Type of machine learning algorithm(s) the group intends to use
 - Research question
 - Sources of bias
 - Dataset(s) to be used
 - At least 3 to 4 citations



Proposal Presentation Rubric

While there is no Proposal Presentation rubric, you are encouraged to reference the [Project Proposal Rubric](#).



Final Presentation (Slides)

The purpose of the Final Presentation is to showcase what you and your group have learned and accomplished throughout the program. Your group will share its presentation with the entire class during a showcase at the end of the program.

Groups should share their Final Presentation material (slides, links, docs, etc.) in a post on Ed STEM within 24 hours of Week 12's Final Showcase.

Final Presentation requirements:

- 10-minute presentation
- Description of the project and how it evolved throughout this program
- Demo of the model (if applicable)
- Evaluation of the model
- At least one data visualization
- Comment on the positive and negative impact of this type of work
- Explain the algorithm used:
 - Name of algorithm(s) used
 - Type of algorithm used (supervised, unsupervised, etc.)
 - Inputs and outputs of the algorithm
- A concise answer to the essential question: *How can the process of creating AI/ML solutions amplify or mitigate bias in the case our group is using?*
- Link to GitHub repository
- Citations:
 - At least 3 to 4 citations related to the topic of interest
 - The source of data included
- Next steps:
 - This can be a description of how the group plans to continue working together or a summary of individual group members' next steps based on their My AI Roadmap assignment



Final Presentation Rubric

No submission, missing names, or lack of citations are graded as a zero. To be [nominated to present at the Student Symposium](#), student groups must score at least 24/30 on this rubric.

RUBRIC Final Presentation					
Criteria	5 Exceeds Requirements	4 At Times Exceeds Requirements	3 Meets Requirement	2 Beginning to Meet Requirements	1 Needs Major Improvement
Project Description	Provides a high-level description of the project and what was being solved or investigated. Comments on how the project changed over the program. Describes at least one challenge faced and solutions implemented.	Provides a high-level description of the project and what was being solved or investigated. Comments on how the project changed over the program.	Provides a high-level description of the project and what was being solved or investigated. Does not mention if/how the project changed over the program.	Describes the project but it is unclear as to what was being solved or investigated.	No project description is included.



Data Visualization	Includes at least three data visualizations that help the viewer understand the data or challenge being addressed.	Includes at least two data visualizations that help the viewer understand the data or challenge being addressed.	Includes at least one data visualization that helps the viewer understand the data or challenge being addressed.	Includes at least one data visualization but it is unclear as to what is being communicated.	No data visualizations were included.
Algorithm	Provides a strong explanation of algorithm(s) used including the type (supervised, unsupervised, etc.), inputs, and outputs. Conveys a solid understanding of the algorithm and its data. Includes pros & cons of selection.	Provides definition(s) of the algorithm(s) used including the type (supervised, unsupervised, etc.), inputs, and outputs. Goes beyond mere recall of definitions and begins to describe the relationship between the algorithm and data.	Provides simple definition(s) of the algorithm(s) used including the type (supervised, unsupervised, etc.), inputs, and outputs. A deep understanding of the algorithm and its relationship to data is not conveyed.	Provides simple definition(s) of the algorithm(s) used including the type used (supervised, unsupervised, etc.), inputs, and outputs. Explanation includes minor errors.	Does not provide a definition(s) of the algorithm(s). OR Explanation includes major errors.



Essential Question	Provides an answer to the essential question grounded in research, discussion, and reflection. Takes into account the complexity of AI's impact. The answer goes well beyond the content covered in the program.	Provides an answer to the essential question grounded in research, discussion, and reflection. Takes into account the complexity of AI's impact.	Provides an answer to the essential question. Begins to take into account the complexity of AI's impact.	Provides an answer to the essential question, however, the answer only emphasizes the positive or the negative aspects of AI/ML.	Does not provide an answer to the essential question.
Next Steps	The next steps are measurable, well-thought-out, and include deadlines.	The next steps are included and are well-thought-out.	The next steps are included but may seem impractical.	The next steps are included but extremely vague. E.g. "I will learn more."	The next steps are not included.
Citations	At least 7 citations related to the topic of interest. Source of data included.	5 to 6 citations related to the topic of interest. Source of data included.	3 to 4 citations related to the topic of interest. Source of data included.	2 citations related to the topic of interest. Source of data included.	Less than 2 citations related to the topic. OR Source of data not included.
Overall Score: ____/30 (Calculate the total score by summing up the scores for each criterion)					



Student Symposium Nomination

This rubric is designed to help instructors evaluate student projects for nomination to the Student Symposium, focusing on executive presence, teamwork, presentation quality, dependability, and overall project appeal.

Purpose and Use: Instructors will use this rubric after students present their projects in Week 12. Nominations should be based on a thorough evaluation of each group, with Student Coaches (SCs) providing input, particularly on dependability and group collaboration.

Evaluation Criteria:

- **Group Engagement:** Observe how well the group collaborates, communicates, and shares responsibilities. Consider fairness by assessing both group dynamics and individual contributions.
- **Technical Skills vs. Presentation:** Strike a balance between technical depth and presentation quality to avoid bias toward groups that present more confidently.
- **Bias Mitigation:** Pay close attention to how the group responds to the Essential Question and address ethical considerations, especially regarding AI/ML bias.

Instructor nominations will reflect these criteria, with SCs providing additional insights where appropriate.



Student Symposium Nomination Rubric

To be considered for nomination, student groups must have scored at least 20/25 on the [Final Presentation Rubric](#). Additionally, student groups must score at least 20/25 on this rubric.

RUBRIC Student Symposium Nomination					
Criteria	5 Exceeds Requirements	4 At Times Exceeds Requirements	3 Meets Requirement	2 Beginning to Meet Requirements	1 Needs Major Improvement
Executive Presence	Group demonstrates exceptional confidence, professionalism, and engagement throughout the presentation.	Group presents with strong confidence and professionalism; minor engagement issues.	Group is generally confident but lacks some professionalism or engagement.	Group shows limited confidence or professionalism; disengaged at times.	Group lacks confidence, professionalism, or engagement, significantly detracting from the presentation.
Group Dynamics & Collaboration	Group shows excellent collaboration, with equal participation and mutual support among all members.	Group shows strong collaboration; most members contribute equally with minor imbalances.	Group members participate adequately but with noticeable imbalances in contributions.	Group collaboration is weak, with some members dominating or disengaging.	Group lacks cohesion, with minimal collaboration or participation from some members.
Presentation Skills	Presentation is highly engaging, well-structured, and clearly	Presentation is clear and well-structured, though transitions	Presentation is functional, though lacks strong engagement,	Presentation is unclear or disorganized, with noticeable issues in	Presentation is disjointed, unclear, or difficult to follow, with poor



	communicated, with seamless transitions between members.	or engagement could be improved.	structure, or clarity at times.	communication or transitions.	communication and structure.
Group Dependability	Group consistently met deadlines, showed full engagement, and worked dependably, with each member contributing equally.	Group met most deadlines, remained engaged, and worked dependably, with minor issues in contribution or engagement.	Group was generally dependable, though some members contributed unevenly or missed a deadline.	Group missed multiple deadlines or exhibited inconsistent engagement and dependability from members.	Group missed most deadlines and showed low engagement, with one or more members not contributing at all.
Project Appeal	Project is highly compelling, innovative, and relevant, with strong potential for real-world impact or further development.	Project is interesting, relevant, and demonstrates strong potential for impact or further development.	Project is solid and relevant, but lacks significant innovation or impact potential.	Project is somewhat relevant but lacks clear appeal or potential for development.	Project lacks relevance, innovation, or appeal, with minimal potential for further development or impact.
Overall Score: ____/25 (Calculate the total score by summing up the scores for each criterion)					



GitHub Page

The purpose of the GitHub Page is to showcase what you and your group have learned and accomplished throughout the program. Your group will share its poster with potential employers during the Student Symposium.

GitHub Page requirements:

- Project Overview:
 - Clear description of the project and its evolution
 - Explanation of algorithm(s) used, including type, inputs, and outputs
 - Evaluation of the model with key metrics
- Visuals & Design:
 - At least one effective data visualization
 - [Visually appealing, professional layout](#)
- Impact & Bias:
 - Comment on the positive and negative societal impact
 - Answer how AI/ML solutions can amplify or mitigate bias in the case study
- Citations & Documentation:
 - 3-4 relevant citations and data sources included
 - Link to GitHub repository with code and documentation
- Next Steps
 - Plan for continued group work or individual steps



GitHub Page Rubric

No submission, missing names, or lack of citations are graded as a zero.

I. Overall Rubric

RUBRIC I GitHub Page					
Criteria	5 Exceeds Requirements	4 At Times Exceeds Requirements	3 Meets Requirement	2 Beginning to Meet Requirements	1 Needs Major Improvement
Grammar & Punctuation	Writing is free from grammar and punctuation errors and flows well throughout the entire project.	Minor grammatical or punctuation errors that do not interfere with understanding.	Some grammatical errors and inconsistent punctuation, but the meaning remains clear.	Frequent errors in grammar and punctuation that occasionally obscure meaning.	Significant grammar and punctuation issues that make the project difficult to understand.
Clarity & Structure	Ideas are presented logically, with a clear and cohesive structure. The purpose of the project is easy to follow.	Ideas are generally clear, with minor structural issues. The project is mostly easy to follow.	The structure is somewhat disjointed, and the project's purpose requires effort to discern.	Poor structure, with ideas presented confusingly. The purpose of the project is unclear.	Lacks structure, and the project is difficult to follow. The purpose is missing or entirely unclear.



Presentation of Visuals & Supporting Materials	Visuals (e.g., graphs, charts, images) are clear, well-labeled, and enhance understanding. They are integrated smoothly into the project.	Visuals are mostly clear and relevant but may have minor issues in labeling or integration.	Visuals are present but may be unclear, improperly labeled, or not fully relevant to the content.	Visuals are confusing, poorly labeled, or do not add value to the project.	No visuals or supporting materials, or they are entirely irrelevant or confusing.
Technical Depth & Analysis	The project demonstrates a deep understanding of the technical aspects, with clear and accurate analysis of the problem. All relevant details (e.g., algorithms, datasets) are presented.	The project demonstrates good technical understanding, though some details may be missing or unclear.	The project shows a basic level of technical understanding but lacks depth in analysis or important details.	The project shows limited technical understanding and omits key elements of the analysis.	No technical understanding is demonstrated, and the analysis is absent or severely flawed.
Model Selection & Evaluation	The project clearly explains the model(s) used, why they were selected, and how they were evaluated. The evaluation metrics are appropriate and well-interpreted.	The project explains the model(s) and their evaluation but may lack depth or clarity in the justification and interpretation of metrics.	Basic explanation of the model(s) and evaluation metrics, but important details are missing or unclear.	Limited explanation of the model(s) and evaluation, with little connection between the model choice and results.	No explanation of model selection or evaluation, or significant errors are present.



Use of Data & Results	Data is well-prepared, relevant, and appropriately used in the analysis. Results are clearly presented with meaningful interpretation and discussion.	Data is mostly relevant and prepared, though minor issues may exist in its use. Results are presented but could be interpreted more deeply.	Data is used in the analysis, but key details are missing, and results are not fully explored.	Data is poorly used or irrelevant, and the results are unclear or not properly interpreted.	No relevant data is used, or the results are incorrect or missing.
Overall Score: ____/30 (Calculate the total score by summing up the scores for each criterion)					



Peer Feedback Protocol

Throughout the program, you will be providing feedback to your peers on various assignments, including their Project Proposals and Final Presentations. Feedback is most helpful when it is constructive and actionable. One way to do so is to use the C-O-I-N formula: Context, Observation, Impact, and Next.

CONTEXT

Define the situation

"During your presentation this morning..."

OBSERVATION

Describe the behavior

...I noticed that you were focused on the presentation screen..."

IMPACT

Clarify the impact

...It made me feel I didn't have a chance to ask questions or get clarity on parts I didn't understand..."

NEXT

Make a suggestion

...Facing your audience will allow you to gauge when to elaborate and remind you to stop for questions."

The **C.O.I.N. Conversation Model**, adapted from Anna Carroll's *The Feedback Imperative*
<https://www.mindtools.com/a94k5vp/the-coin-conversation-model>



Reflection Post Response Guidelines

At the end of each module, you are expected to share a reflection post in addition to posting a response to one of your classmates' reflection posts. To help you think through possible acceptable responses, please refer to the following options¹:

- Validate the post by sharing your experience and stating how it relates to the course material or the initial post.
- Agree or disagree with the post and explain why you agree or disagree.
- Expand on your classmate's post to demonstrate that you understand the topic.
- Reply to a question posed by one of your classmates or by the Instructor and support your statements with sources from the text.
- Critically evaluate the post and make a suggestion or respectfully point out an area of the assignment that was not addressed.
- Ask a probing question or ask your classmate for clarification or an explanation of a point made in the post.
- Share an insight or something you learned from reading your peer's posting or from your other reading.
- Explain how someone's post helped you understand the material or made you rethink your views.
- Offer an opinion and support it with examples from the text.
- Relate the information in the post to your course assignments and/or research projects.
- Challenge a statement in the post. If you would not let a statement go unchallenged in a classroom discussion, do not let it go unchallenged on the discussion board.
- Tie ideas together: Discuss how the comments other students made relate to your peer's post.
- Be sure to cite your sources.

¹ "Effectively Responding to a Peer's Post Another component"

<https://www.mssu.edu/academics/distance-learning/pdfs/Effectively%20Responding%20to%20a%20Peer.pdf>. Accessed 25 Aug. 2020.



GitHub Repository

Students will create a GitHub repository following AI4ALL's [GitHub Guidelines](#) that will showcase their work and that can be shared publicly.